NH100 REPLACE ENGINE CONTROL UNITS ON 3 GENERATORS
NAVAL MEDICAL CENTER CAMP LEJEUNE
AT
CAMP LEJEUNE, NORTH CAROLINA 28533-0006

1. **GENERAL INTENTION:** It is the declared and acknowledged intention of this solicitation to obtain the supervision, expertise, labor, materials, and equipment to replace 3 Engine Control Units on Emergency Generators at NH100 Naval Hospital to restore compliance to emergency electrical distribution system.

2. **LOCATION:** Building NH100, Camp Lejeune in North Carolina.

3. **CURRENT SITUATION:** The Emergency Power Supply System (EPSS), three (3) generators, controllers, and emergency power supply distribution system are in various stages of failure but currently crippled with immediate action required. One generator is in manual configuration with a temporary Engine Control Unit (ECU) to keep it online and the other two generators locked out due to ECU failure. The ECUs are not repairable and must be replaced. Hospital Operations are, and will, continue to operate in a reduced capacity until Emergency Power capabilities are restored. One generator *CANNOT* maintain mission requirements of code compliant healthcare delivery. Upon unmitigated or uncontrolled loss of electrical utility power, the sole generator will immediately move into load shedding to only support what it can until it runs to failure. This is a high-risk scenario.

   Current state is that all three generator ECUs must be replaced and the ECU replacements that Detroit Deisel provides require the accompanying onboard controllers, custom fabricated wiring harnesses and associated seals, pulse-controlled fuel injectors, voltage regulators, and associated controls to be replaced in conjunction with the ECU replacement in order for them to work together as originally designed. Once installed, these components must be programmed to be integrated into the existing PLC controllers in paralleling switchgear to ensure they synchronize and coordinate power loads and connectivity to the electrical distribution system under current control parameters.

   The contractor shall replace the ECUs on all three EPSS generators and all associated appurtenances in order to restore automatic EPSS operations. Once all components are installed, the contractor shall hire a verified Commissioning Agent and perform a complete generator system acceptance test in accordance with National Fire Protection Association NFPA 110. The NFPA 110 acceptance test shall be witnessed by DHA and the facility manager and final acceptance will be determined by DHA. Upon completion, the contractor shall provide documented training to the maintenance staff, all Operations and Maintenance (O & M) manuals, and troubleshoot if compliance is not achieved.

   Because MTU/Detroit Diesel does not publish public part numbers for custom-engineered control retrofits, bidders shall utilize the unique Engine Serial Numbers (ESNs) and generator data provided below as their technical keys to request and obtain

firm, custom-engineered upfit kit configurations and pricing directly from the authorized MTU standby power generation distributor.

The following list contains model number, serial number, and nameplate rating information for each diesel prime mover and generator:

"Generator 1": Engine Manufacture Date: January 2006
Engine Control Unit Model #: 5275307654
Engine Control Unit Serial #: 070531701
Prime Mover/Engine Model #: R1637M36
Prime Mover/Engine Serial #: 5362003429
Prime Mover/Engine Power: 1493 bhp (kw)
Prime Mover/Engine Speed: 1800 rpm
Generator Manufacture Date: February 2006
Generator Model #: 1000DSEB
Generator Serial #: 2085473
Generator Spec.: GM30408-GA17S
Service Duty: Prime
Generator Hertz: 60 HZ
Generator RPM: 1800 rpm
Generator Voltage: 480/277 Volt, 3 Phase
Generator Amperage: 1368 Amps
Generator KW Rating: 910 KW
Generator Power Factor: 0.8

"Generator 2": Engine Manufacture Date: January 2006
Engine Control Unit Model # 5275307654
Engine Control Unit Serial # 073539995
Prime Mover/Engine Model #: R1637M36
Prime Mover/Engine Serial #: 5362003428
Prime Mover/Engine Power: 1493 bhp (kw)
Prime Mover/Engine Speed: 1800 rpm
Generator Manufacture Date: February 2006
Generator Model #: 1000DSEB
Generator Serial #: 2085471
Generator Spec.: GM30408-GA17S
Service Duty: Prime
Generator Hertz: 60 HZ
Generator RPM: 1800 rpm
Generator Voltage: 480/277 Volt, 3 Phase
Generator Amperage: 1368 Amps
Generator KW Rating: 910 KW
Generator Power Factor: 0.8

"Generator 3": Engine Manufacture Date: December 2005
Engine Control Unit Model # 5275307654
Engine Control Unit Serial # 070539008
Prime Mover/Engine Model #: R1637M36
Prime Mover/Engine Serial #: 5362003410
Prime Mover/Engine Power: 1493 bhp (kw)
Prime Mover/Engine Speed: 1800 rpm
Generator Manufacture Date: February 2006
Generator Model #: 1000DSEB
Generator Serial #: 2085472
Generator Spec.: GM30408-GA17S
Service Duty: Prime
Generator Hertz: 60 HZ
Generator RPM: 1800 rpm
Generator Voltage: 480/277 Volt, 3 Phase
Generator Amperage: 1368 Amps
Generator KW Rating: 910 KW
Generator Power Factor: 0.8

4. DETAILED REQUIREMENTS: The Contractor shall:

a. Make a site visit and take such other steps as may be reasonably necessary to ascertain the nature and location of the work and general local conditions that could affect the work or the cost thereof and should perform an inspection of each repair item, taking their own measurements and developing their own material requirements in accordance with the specifications. Failure to do so will not relieve the contractor from the responsibility of estimating properly the difficulty or cost of successfully performing the work. Adhere to all specified information within these pages.

b. Basis of Design:

Due to the severity and direct mission degradation and impact to NMCCL's mission set of patient care and healthcare delivery, all mechanical and electrical upfit work on the prime movers, including ECU programming and proprietary firmware flashing, shall only be performed by factory-certified technicians from an authorized representative of Detroit Diesel/MTU (e.g., Curtis Power Solutions). Proof of OEM certification shall be provided in the technical submittal package.

Provide a minimum of 2 onsite OEM-certified technicians concurrently to accomplish the removal of the 3 existing engine control units and associated generator controls, wire-for-wire reconfiguration of the engine control unit upfit kits, installation of new OEM pulse-controlled fuel injectors and custom-fabricated wiring harnesses, removal of existing generator voltage regulators, and installation and wire termination of new Marathon DVR 2400 digital voltage

regulators.

The Prime Contractor shall not permit unauthorized electricians or third-party integrators to modify existing RussElectric PLC source code. The Contractor shall subcontract directly with the installed paralleling switchgear manufacturer (RussElectric/Siemens Field Service) to provide a factory-certified field service engineer on-site during integration and testing. The MTU technician shall land and verify all engine and regulator control signals up to the generator-side terminal blocks, and the RussElectric field engineer shall verify switchgear-side terminations, execute necessary parameter updates within the RussElectric PLC/master control cubicles, and co-witness acceptance testing with DHA to verify that all repairs to Engine Control Units, Generator Controls, and paralleling sequences are satisfactorily executed within requirements from UFC, NFPA, and The Joint Commission.

The work shall be phased to accomplish all work on locally identified engine "one" (1) and "three" (3) as phase one of the project. The first phase replacement of engine control units, controls kits, voltage regulators, and injectors shall be inspected by the Contracting Officer's Representative (COR) prior to commencement of work in the second phase. At the completion of phase one work, a "no load" operational test shall commence to verify engine outputs are satisfactory to the required parameters and codes. The second phase shall consist of replacement of the engine control unit, controls kit, voltage regulator, and injectors on engine two (2). Phase 2 work shall be inspected by the Contracting Officer's Representative (COR). *Under no circumstances shall Generator 2 be decommissioned or taken out of service until both Generator 1 and Generator 3 have been fully upgraded, operationally tested, and certified by all approving authorities for active service.*

The Contractor shall hire an independent, verified Commissioning Agent that has validated experience restoring compliance to legally required healthcare emergency power distribution systems. The Commissioning Agent shall ensure all operational "load tests" are completed to required standards of NFPA 99, NFPA 110, The Joint Commission, and UFC 4-510-01. Prior to placing upgraded generators onto the live hospital bus, the Prime Contractor shall perform full-capacity acceptance testing utilizing temporary exterior resistive/reactive load banks. The Prime Contractor is wholly responsible for providing, delivering, rigging, connecting, and disconnecting the temporary resistive/reactive load banks and all required temporary feeder cables rated for full generator output (910 KW / 0.8 PF per unit). The Government (NMCCL) shall provide diesel fuel from the existing bulk day-tank/storage system for all on-site testing. The Contractor shall coordinate runtime schedules with NMCCL FACMAN 72 hours prior to testing, so equipment/plant status is communicated to and from NMCCL Command/Directorates for healthcare delivery.

The Contractor shall coordinate with NMCCL FACMAN to conduct an integrated dynamic "load test" utilizing designed and connected hospital facility loads and automatic transfer switches to ensure all respective paralleling gear programming reacts as designed. Physical transfer of hospital building loads shall be operated by NMCCL FACMAN staff, witnessed by the Contractor, Commissioning Agent, MTU technician, RussElectric field engineer, COR, and DHA, and accepted by the Government. Upon successful dynamic "load testing", the Contractor shall perform a resistive/reactive load bank "load test" as required by all codes to restore compliance to emergency power distribution system.

Upon completion, the Contractor shall provide a maximum of four (4) hours of documented hands-on training to the maintenance staff, three (3) hardcopies and one (1) electronic PDF copy of all updated Operations and Maintenance (O & M) manuals, as-built wiring diagrams, and final NFPA 110 commissioning reports, and troubleshoot until full compliance is achieved.

Prior to construction, take all precautions necessary to protect existing property. Existing property damaged, by work on this project, shall be returned to original condition or better. Note: This shall include damage to grounds, turf and grade by heavy equipment such as lifts. Damaged turf and grade shall be restored to existing contours and stabilized with centipede sod.

5. **SPECIAL REQUIREMENTS:** The Contractor shall be required to:

a. Accomplish all work in a professional manner; taking in consideration the work performed is adjacent to a medical facility containing a multitude of professional staff and the patients. The contractor shall minimize the impact imposed on those persons and will show courtesy and consideration throughout the course of this contract.

b. Coordinate work schedule with the Contracting Officer prior to the commencement of work.

c. Building will be in operation during the duration of this contract.

d. Have 100% of all OEM materials, custom harnesses, injectors, and voltage regulators physically on-site and inventoried with the COR prior to commencement of on-site physical installation work.

e. Install all materials in accordance with all manufacturer's instructions.

f. Coordinate all work and schedules with the Contracting Officer's designated representative and NMCCL FACMAN. All work will happen during the hours of 0700-1900 Monday through Friday. Any work performed outside of these hours must be pre-approved.

g. Prosecute the work diligently.

h. Because specialty Detroit Diesel/MTU ECUs, custom-fabricated wiring harnesses, and Marathon voltage regulators have variable factory lead times that are not published until an active order is placed, the Period of Performance shall be divided into two phases: Phase I (Submittal and Procurement Phase) requires the Contractor to submit all technical submittals within fifteen (15) calendar days of contract award, immediately order all OEM materials upon approval, and provide bi-weekly factory tracking updates to the COR and customer (NMCCL FACMAN); this procurement phase shall not count against the physical construction window. Phase II (Physical Construction and Commissioning Phase) requires the Contractor to complete all on-site installation, integration, testing, commissioning, and closeout within sixty (60) calendar days from verified on-site receipt and joint COR inventory of 100% of manufacturer materials.

i. All material associated with the repair, replacement, or removal is the property of the contractor and shall be removed from the base and discarded in an approved landfill. The building's trash dumpster is not to be used to dispose of construction debris.

j. All North Carolina Building Codes, Life Safety Codes, Fire Codes, Statutes, NEC, OSHA Safety Codes, UFC 4-510-01, UFC 3-540-07, and the U.S. Army Corps of Engineer's Safety and Health Requirements Manual, EM 385-1-1 shall be strictly enforced. An accident prevention plan in accordance with EM 385-1-1 is required. The contractor is responsible for any required permits.

k. All contractor/subcontractor vehicles parked outside of construction boundaries will be parked in the designated parking lots.

l. NMCCL Staff WILL NOT be responsible for sponsoring any one day passes for access to the installation. This will be the sole responsibility of the contractor.

m. The Contractor shall be working in an existing building which is occupied. Do not enter the buildings without prior approval of the Contracting Officer.

n. Remove construction waste products from the site by covering and transporting them to dumpster. Waste is not to be transported through patient/staff areas without the approval of the NMCCL Project Manager and affected Department Director.

o. The use of tobacco products (electronic cigarette, smoking and smokeless) is not authorized on the premises.